Lab 4:
Stretch Moves Around

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Lab 4 Goals

In this lab, you'll be asked to map the AI Maker Space and write Python code to have the robot navigate to different locations in the AI Maker Space.

Part #1

Build a map of the AI Maker Space using Stretch’s SLAM implementation.

Part #2

Use your map to navigate to four different base poses within the AI Maker Space.

What you will submit

1. Part #1: Your map files saved from SLAM.
    - `${HELLO_FLEET_PATH}/maps/<map_name>.yaml`
    - `${HELLO_FLEET_PATH}/maps/<map_name>.pgm`
2. Your completed code file (.py) for Part #2 autonomous navigation.
    - `stretch_navigation.py`
3. Video (.mp4 or .mov) from a smartphone showing the robot navigating to 4 different locations within the AI Maker Space.
4. Video (.mp4 or .mov) from the robot's head-mounted RealSense camera as it is navigating the AI Maker Space.

Lab 4

- **Wednesday, March 25th**

Reminder: How to connect into the robot

- We have two robots — 3159 and 3160. These numbers are on the back of the robot near the on/off switch.

- Username: hello-robot
- Password: 16762cmu!

- stretch-se3-3159:
    - **SSH**: ssh -X hello-robot@stretch3159.wifi.local.cmu.edu
    - **Anydesk**: 1943010008

- stretch-se3-3160:
    - **SSH**: ssh -X hello-robot@stretch3160.wifi.local.cmu.edu
    - **Anydesk**: 1424568870

Lab 4

Lecture 7 slides provide details needed to complete both parts #1 and #2.

Part #1: Building your map

- Run `offline_mapping.launch.py` to build a map of the AI Maker Space.
- You can use the Xbox controller or the keyboard to move the robot.
- When saving your map, you should give it a filename that is unique to your team, so you don’t accidentally overwrite another team’s map.

Part #2: Autonomous Navigation

- Write a Python script that has Stretch autonomously navigate to 4 different base poses (position + orientation), not including the start pose in the AI Maker Space and record a video from the robot's head-mounted camera as it moves throughout the AI Maker Space.
- Each pose must be at least 2 meters away from each other and you should not reuse a pose that is similar to a prior pose.

- You may find the below script helpful to start:
    - https://github.com/hello-robot/stretch_ros2/blob/humble/stretch_nav2/stretch_nav2/simple_commander_demo.py
- Note: this script only matches position, your script must also align the robot to specific orientations (yaw) of the robot. You will also need to save image frames from the robot's head camera into a video as the robot is navigating.